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RVCOLLEGE OF ENGINEERING®*(Autonomous Institution affiliated to VTU)***III Semester B.E. Examinations****Department of Industrial Engineering and Management****Model Question Paper****Course: DIGITAL METROLOGY****(2022 – Scheme)****Time:03 Hrs****MaximumMarks:100****Instructions to candidates:**

1. Answer all questions from PartA. PartA questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from PartB. In PartB question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6,7 and 8 and 9 and 10.

PART-A

1	1.1	List any two desirable static and dynamic characteristics of measuring instruments.	02	L2	CO1
	1.2	Mention the working principle of an Inductive sensor.	02	L1	CO2
	1.3	Categorize the two applications of a Tactile sensor and analyze the principles underlying the use of tactile sensors in each application	02	L3	CO2
	1.4	Identify the two techniques used for converting Digital Signals into Analog signals	02	L2	CO2
	1.5	Compare and contrast the Hole basis system and shaft basis system in terms of their application in geometric dimensioning and tolerancing (GD&T)	02	L3	CO1
	1.6	Define Maximum clearance and Fundamental deviation	02	L1	CO1
	1.7	Evaluate the importance of optical flats in measuring surface finish, considering their role in precision measurement techniques	02	L3	CO3
	1.8	Evaluate the significance of determining a Ten-point Height Average Value in surface metrology.	02	L3	CO3
	1.9	Classify the primary components of a Coordinate Measuring Machine (CMM) according to their functions	02	L2	CO4
	1.10	Classify the different geometrical checks performed on machine tools using LASER technology and analyze the principles behind each check	02	L3	CO4

PART-B

2	a	Describe briefly the three stages of a generalized measurement system with block diagram. Provide two specific examples of measurement systems and identify how these examples demonstrate the three stages of measurement according to the categorized stages mentioned.	08	L3	CO1
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	b	Illustrate the operational principles of two distinct resistive transducers, namely the strain gauge and the thermistor, and compare their applications in real-world scenarios	08	L3	CO2
3	a	Explain the functioning of tactile sensors detailing its basic operation and how it detects contact with surfaces. Evaluate the advantages and limitations of this sensor in different scenarios, such as robotics, medical devices, touch-sensitive electronics, and industrial monitoring.	08	L4	CO2
	b	Describe the operational principles of Potentiometer Sensors based on resistance change. Explain how these sensors function to measure position, displacement, or angle by varying resistance. Provide examples of their applications in volume control knobs, servo motors, joysticks, and industrial position feedback systems	08	L4	CO2
		OR			
4	a	Describe the operational principles of a tachogenerator and its role in measuring rotational speed. Evaluate the role of tachogenerators in measuring and monitoring rotational speed in automotive systems and automation/robotics.	08	L4	CO2
	b	Enumerate briefly on the classification of sensors based on different criteria and explain the working principle of Hall effect sensors. Discuss how Hall effect sensors generate a voltage difference in response to a magnetic field and how this principle is applied in various applications such as proximity sensing, position sensing, and speed measurement	08	L4	CO2
5	a	Compare and contrast the characteristics, advantages, and applications of different types of fits in engineering assemblies. Also evaluate their suitability for specific assembly requirements.	08	L4	CO1
	b	Determine the actual dimension to be provided for a shaft and hole of 90 mm size for h ₈ E ₉ type of fit. i. Size 90 mm falls in the diameter step of 80mm-100mm. ii. Value of tolerances for IT8 and IT9 are 25i and 40i. iii. Value of fundamental deviation for 'e' type shaft is 11D ^{0.41} . Also mention the type of fit. Represent the dimensions graphically.	08	L4	CO1
		OR			
6	a	Explain briefly the Hole basis system and shaft basis system in geometric dimensioning and tolerancing (GD&T). Analyze the principles behind each system, evaluate their advantages and limitations, and assess their effectiveness in ensuring proper fit and functionality in mechanical components, such as engine parts or aerospace components.	08	L4	CO1
	b	Enumerate on the concepts of interchangeability and selective assembly in manufacturing processes. How do these principles contribute to efficient production and product quality? Provide examples to illustrate your explanation.	08	L3	CO1
7	a	Apply the principle of interference to demonstrate the functioning of an autocollimator? Furthermore, discuss how autocollimators are practically utilized in manufacturing settings to assess the straightness of components.	08	L4	CO3
	b	Define Crest, Flank and Pitch and schematically represent on a screw thread. Explain how the major diameter of a screw thread can be measured using Bench micrometer.	08	L3	CO3
		OR			

8	a	Discuss the Tomlinson Surface Meter, its construction, principles of operation, and applications in surface finish measurement. Explain how this specialized instrument contributes to accurate surface finish analysis in engineering and manufacturing processes, highlighting its advantages and limitations.	08	L4	CO3
	b	Define Pitch Circle diameter, addendum and dedendum and represent on a gear profile. Assess the effectiveness of using vernier calipers for gear tooth thickness measurement compared to other measurement methods, and discuss potential sources of error in gear measurements..	08	L3	CO3
9	a	Explain briefly the 4 main elements of a Coordinate Measuring Machine (CMM). Evaluate how each component contributes to the accurate measurement and inspection capabilities of the CMM in various industrial applications.	08	L3	CO4
	b	How does Coordinate Measuring Machine (CMM) integration with Computer-Aided Manufacturing (CAM) enhance quality control processes in manufacturing? Discuss with examples.	08	L3	CO4
		OR			
10	a	Describe the key components of a laser interferometry system in detail, focusing on its different elements. Explain the function of each component and how they collectively enable precise distance measurements and displacement sensing through interference phenomena.	08	L3	CO4
	b	Analyze the components and functions of a control system integrated with a data conversion system. Provide a diagram illustrating signal flow and conversion between analog and digital formats.	08	L3	CO4

.....Q.P.Code (For office use only)

Course Code: IM235AI	Course Title: Digital Metrology
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PART-A

Q.No	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	1.10
B T	2	1	3	2	3	1	3	3	2	3
COs	1	2	2	2	1	1	3	3	4	4
Q No	1.11	1.12	1.13	1.14	1.15	1.16	1.17	1.18	1.19	1.20
B T										
COs										

PART-B

Question No		B T Levels	Cos addressed		Question No		BT Levels	Cos addressed
2	a	3	1		3	a	4	2
	b	3	2			b	4	2
	c					c		
	d					d		
4	a	4	2		5	a	4	1
	b	4	2			b	4	1
	c					c		
	d					d		
6	a	4	1		7	a	4	3
	b	3	1			b	3	3
	c					c		
	d					d		
8	a	4	3		9	a	3	4
	b	3	3			b	3	4
	c					c		
	d					d		
10	a	3	4		11	a		
	b	3	4			b		
	c					c		
	d					d		

Signature of Scrutinizer:

Signature of Chairperson:

Name:

Name:

